

4.3.2. Operational Discovery

For Matter Nodes that have already been commissioned onto a Matter Fabric, run-time dynamic discovery of operational Matter Nodes is used, rather than assuming a fixed unchanging IPv6 address and port for the lifetime of the product. This is done to allow for greater flexibility, so that the underlying IPv6 network can grow and evolve over time as needed without breaking Matter functionality. This is the same reason that other networked consumer electronics products do not assume a single fixed unchanging IP address for the lifetime of the product [RFC 5505].

4.3.2.1. Operational Instance Name

For Matter operational discovery the DNS-SD instance name is constructed from a 64-bit [compressed Fabric identifier](#), and a 64-bit Node identifier, as assigned by the commissioner, each expressed as a fixed-length sixteen-character hexadecimal string, encoded as ASCII (UTF-8) text using capital letters, separated by a hyphen. For example, a Matter Node with Matter compressed fabric identifier **2906-C908-D115-D362** and Matter Node identifier **8FC7-7724-01CD-0696** has Matter operational discovery DNS-SD instance name **2906C908D115D362-8FC7772401CD0696**.

The Matter operational discovery DNS-SD instance name needs to be unique within the namespace of the local network (the **.local** link-local namespace of the Ethernet and Wi-Fi links [RFC 6762], or the unicast domain selected by the Thread Border Router for devices on the Thread mesh). This uniqueness is assumed to be guaranteed by appropriate selection of a unique Matter fabric identifier and unique Node identifier within that Matter fabric.

4.3.2.2. Compressed Fabric Identifier

In order to reduce the very large size of a full [Fabric Reference](#) which would need to be used as the scoping construct in the [instance name](#), a 64-bit compressed version of the full Fabric Reference SHALL be used. The computation of the Compressed Fabric Identifier SHALL be as follows:

```
byte CompressedFabricInfo[16] = /* "CompressedFabric" */
    {0x43, 0x6f, 0x6d, 0x70, 0x72, 0x65, 0x73, 0x73,
     0x65, 0x64, 0x46, 0x61, 0x62, 0x72, 0x69, 0x63}

CompressedFabricIdentifier =
    Crypto_KDF(
        inputKey := TargetOperationalRootPublicKey,
        salt := TargetOperationalFabricID,
        info := CompressedFabricInfo,
        len := 64)
```

Where:

- **TargetOperationalRootPublicKey** is the raw uncompressed elliptical curve public key of the root certificate for the advertised Node's [Operational Certificate](#) chain, without any format marker prefix byte (i.e. after removing the first byte of the **ec-pub-key** field in the Operational Certificate's root).